Modeling

Concept	Formula	Concept	Formula
Variance, σ_x^2	$\sigma_x^2 = \frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^2$	Correlation, r	$r = \frac{1}{n}\sum_{i=1}^{n}\frac{x_i - \bar{x}}{\sigma_x}\frac{y_i - \bar{y}}{\sigma_y}$
L_1 loss	$L_1(y, \hat{y}) = \lvert y - \hat{y}\rvert$	Simple linear regression (SLR) prediction	$\hat{y} = \theta_0 + \theta_1 x$
L_2 loss	$L_2(y, \hat{y}) = (y - \hat{y})^2$	SLR slope and intercept	$\hat{\theta}_1 = r\frac{\sigma_y}{\sigma_x}$ $\hat{\theta}_0 = \bar{y} - \hat{\theta}_1\bar{x}$
Empirical risk with loss L	$R(\theta) = \frac{1}{n}\sum_{i=1}^{n} L(y_i, \hat{y}_i)$		

Multiple Linear Regression Formulas

Concept	Formula	Concept	Formula
Mean squared error (MSE)	$R(\theta) = \frac{1}{n}\lVert Y - X\theta\rVert_2^2$	Normal equation	$X^T X\hat{\theta} = X^T Y$
Ordinary least squares (OLS) estimate (if X is full rank)	$\hat{\theta} = (X^T X)^{-1} X^T Y$	Multiple R^2 (coefficient of determination)	$R^2 = \frac{\text{variance of fitted values}}{\text{variance of } \ y}$
Ridge regression (L_2 regularization)	$\frac{1}{n}\lVert Y - X\theta\rVert_2^2 + \lambda\lVert\theta\rVert_2^2$	Squared L_2 norm of $\theta \in \mathbb{R}^d$	$\lVert\theta\rVert_2^2 = \sum_{j=1}^{d}\theta_j^2$
Ridge estimate (closed form)	$\hat{\theta}_{\text{ridge}} = (X^T X + n\lambda I)^{-1} X^T Y$	L_1 norm of $\theta \in \mathbb{R}^d$	$\lVert\theta\rVert_1 = \sum_{j=1}^{d}\lvert\theta_j\rvert$
LASSO regression (L_1 regularization)	$\frac{1}{n}\lVert Y - X\theta\rVert_2^2 + \lambda\lVert\theta\rVert_1$		

Gradient Descent

For a learning rate α, the gradient update step is:

$$\theta^{(t+1)} = \theta^{(t)} - \alpha\nabla_\theta L\big(\theta^{(t)}\big)$$

Here $\nabla_\theta L\big(\theta^{(t)}\big)$ is the partial derivative/gradient of L with respect to θ, evaluated at $\theta^{(t)}$.